

Hibernate Envers Auditing — 2-Page Practical Guide

Purpose

Hibernate Envers is a Hibernate ORM extension for entity auditing/versioning. It records historical snapshots whenever audited entities are changed, allowing an application to answer what changed and retrieve an entity as it existed at a particular revision. A revision represents the audited changes committed in a transaction. Envers integrates with Hibernate; it is not a separate database or a replacement for Hibernate/JPA. [Hibernate Envers documentation]

1. Library / Dependency

```
<dependency>
    <groupId>org.hibernate.orm</groupId>
    <artifactId>hibernate-envers</artifactId>
</dependency>
```

Use the Envers version compatible with the Hibernate ORM version managed by your Spring Boot release. With current Hibernate ORM, Envers is the auditing/versioning module. Adding it to the classpath is sufficient for automatic Envers integration; explicit legacy listener configuration is normally not required. [Hibernate official documentation]

2. Configuration

```
spring.jpa.hibernate.ddl-auto=none
```

Because this example intentionally uses manually maintained SQL, keep Hibernate DDL disabled. Envers can generate audit tables when Hibernate schema generation is enabled, but that is not used here. In a production project, Flyway/Liquibase can own the schema while Hibernate validates/maps it.

3. Example Entity

```
@Entity
@Audited
@Table(name = "requests")
public class Request {
    @Id
    @Column(name = "id", length = 36, nullable = false)
    private String id;
    private Long employeeId;
    private LocalDate startDate;
    private LocalDate endDate;
    @Enumerated(EnumType.STRING)
    private RequestStatus status;
    private Double totalHours;
    private Double totalDays;
}
```

The key annotation is `org.hibernate.envers.Audited`. `@Audited` on the class audits the persistent properties of the entity. Alternatively, `@Audited` can be placed on selected fields/properties when only specific attributes should be audited. [Hibernate Envers documentation]

4. What @Audited Does

- INSERT: creates an audit revision with revision type 0 (ADD).
- UPDATE: creates another historical row with revision type 1 (MOD).
- DELETE: creates a deletion marker with revision type 2 (DEL).
- The audit table normally uses the original entity ID + revision number as its primary key.
- Historical data is stored in a separate <entity-table>_AUD table; the current entity remains in the normal table.
- Audit history is committed with the transaction, so the audit record corresponds to the entity change.

5. Envers Tables / Data Model

For Request, the normal and audit tables are conceptually:

```
requests      -> current state
requests_aud  -> historical states
revinfo       -> global revision metadata
```

The audit table contains the original entity ID, revision number (REV), revision type (REVTTYPE), and audited fields. The revision table identifies the revision and its timestamp. By default Envers commonly uses the REVINFO/revision mapping; the exact generated column names must match the Hibernate/Envers version and configuration.

Hibernate Envers — Database, Querying & Practical Setup

6. Manual MySQL DDL for This Example

Since `spring.jpa.hibernate.ddl-auto=none`, create the schema yourself. The following matches the default revision SQL shown by the user's running application, where Hibernate inserts into `revinfo(revtstamp)`.

```
CREATE TABLE requests (
    id VARCHAR(36) NOT NULL,
    employee_id BIGINT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    status VARCHAR(30) NOT NULL,
    total_hours DOUBLE,
    total_days DOUBLE,
    PRIMARY KEY (id)
);

CREATE TABLE revinfo (
    rev INT NOT NULL AUTO_INCREMENT,
    revtstamp BIGINT NOT NULL,
    PRIMARY KEY (rev)
);

CREATE TABLE requests_aud (
    id VARCHAR(36) NOT NULL,
    rev INT NOT NULL,
    revtype TINYINT NOT NULL,
    employee_id BIGINT,
    start_date DATE,
    end_date DATE,
    status VARCHAR(30),
    total_hours DOUBLE,
    total_days DOUBLE,
    PRIMARY KEY (id, rev),
    CONSTRAINT fk_requests_aud_rev
    FOREIGN KEY (rev) REFERENCES revinfo(rev)
);
```

Important: Envers' revision timestamp column must match the revision entity mapping. If Hibernate logs ``insert into revinfo (revtstamp) values (?)``, the table must contain ``revtstamp``. A manually created ``rev_timestamp`` column alone will cause MySQL error 1054: Unknown column 'revtstamp'. Either create ``revtstamp`` or explicitly configure a custom revision entity/column mapping.

7. Typical Audit Flow

```
requestRepository.save(request);
↓
Hibernate persists requests
↓
Envers creates revision in revinfo
↓
Envers writes snapshot to requests_aud
↓
Transaction commits
```

Example: PENDING → APPROVED produces two audit snapshots: the initial ADD revision and a later MOD revision. A delete produces a DEL revision marker. This allows historical reconstruction of the entity at a revision.

8. Reading Audit History

```
@PersistenceContext
private EntityManager entityManager;

AuditReader reader =
    AuditReaderFactory.get(entityManager);

List<Number> revisions =
    reader.getRevisions(Request.class, requestId);

Request oldRequest =
    reader.find(Request.class, requestId, revisions.get(0));
```

`AuditReader` can retrieve an entity at a specific revision and list revisions for an entity. This is the main API for reading Envers history. [Hibernate Quickstart / Envers documentation]

9. Selective / Advanced Auditing

- Class-level `@Audited`: audit the entity's persistent properties.
- Field/property-level `@Audited`: audit only selected properties.
- `@AuditTable`: customize the audit table name.
- Relations can also be audited; Envers supports historical association information with appropriate mapping/configuration.
- A custom revision entity can add username, IP address, application/service, correlation ID, or other revision metadata.

- AuditReader supports historical snapshots and revision-oriented queries.

10. Recommended Spring Boot Structure

```
Request.java           -> @Entity + @Audited
RequestRepository.java -> normal JPA repository
RequestService.java    -> transactional business logic
V1__create_request.sql -> requests table
V2__create_envers.sql  -> revinfo + requests_aud
application.properties -> ddl-auto=none
```

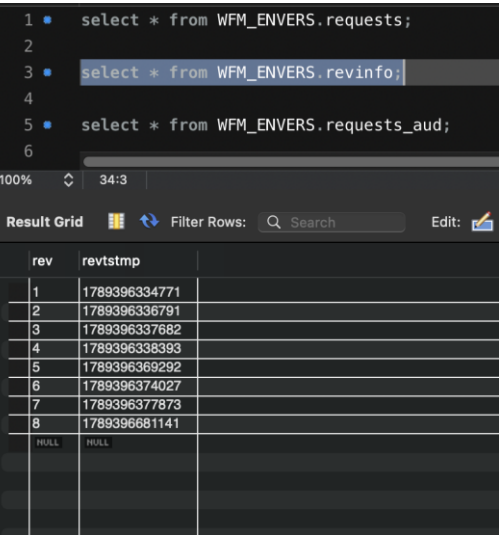
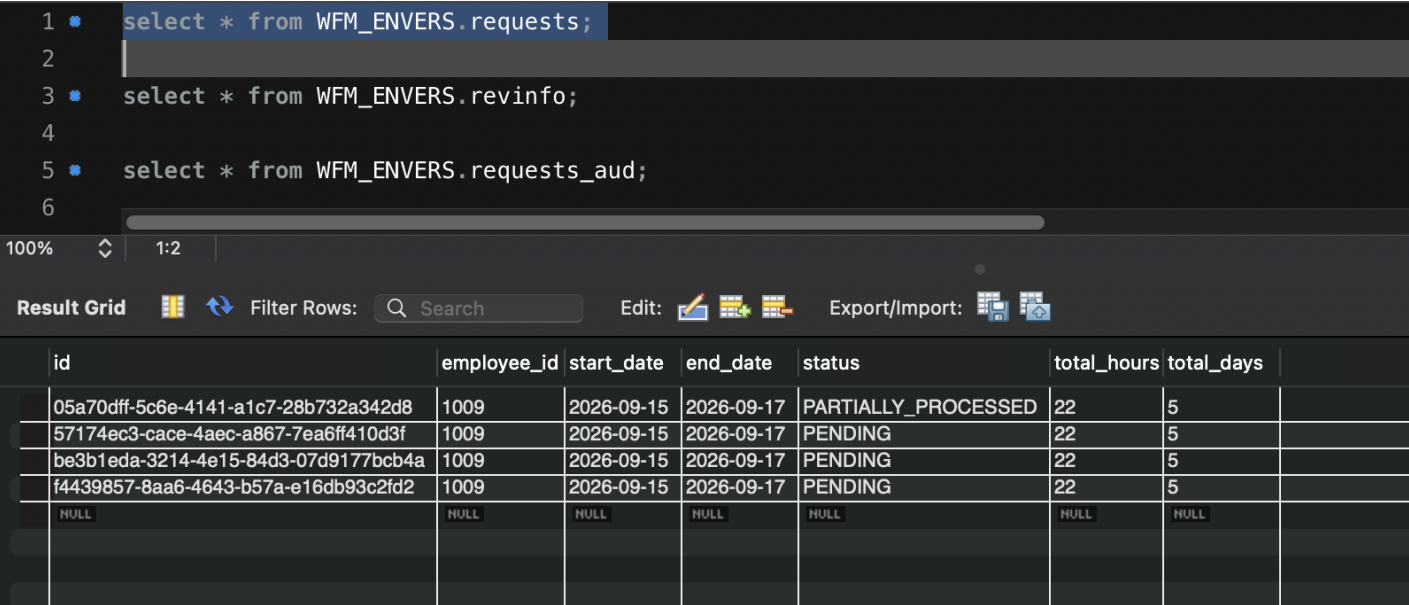
Key Takeaways

- Envers = Hibernate's entity auditing/versioning extension.
- Dependency: hibernate-envers; annotate audited entities with @Audited.
- Audit tables normally use the _AUD suffix and a revision table stores revision metadata.
- REVTYPE: 0 = ADD, 1 = MOD, 2 = DEL.
- Use AuditReader/AuditReaderFactory to retrieve historical versions.
- If SQL is managed manually, Hibernate DDL should remain disabled and every Envers column must exactly match its mapping.
- For production schema management, use Flyway or Liquibase rather than relying on Hibernate DDL generation.

References

Hibernate ORM — Envers: hibernate.org/orm/envers/ | Hibernate ORM Quickstart — Tutorial Using Envers | Hibernate ORM User Guide — Envers schema and auditing. Information checked against current Hibernate documentation; Hibernate ORM 7.4 is listed as the latest stable series in the official documentation as of September 2026.

Images:



```

1 select * from WFM_ENVERS.requests;
2
3 select * from WFM_ENVERS.revinfo;
4
5 select * from WFM_ENVERS.requests_aud;
6

```

100% 1:6

Result Grid Filter Rows: Search Edit: Export/Import:

	id	rev	revtype	employee_id	start_date	end_date	status	total_hours	total_days	
	05a70dff-5c6e-4141-a1c7-28b732a342d8	4	0	1009	2026-09-15	2026-09-17	PENDING	22	5	
	05a70dff-5c6e-4141-a1c7-28b732a342d8	5	1	1009	2026-09-15	2026-09-17	APPROVED	22	5	
	05a70dff-5c6e-4141-a1c7-28b732a342d8	6	1	1009	2026-09-15	2026-09-17	PROCESSED	22	5	
	05a70dff-5c6e-4141-a1c7-28b732a342d8	7	1	1009	2026-09-15	2026-09-17	REJECTED	22	5	
	05a70dff-5c6e-4141-a1c7-28b732a342d8	8	1	1009	2026-09-15	2026-09-17	PARTIALLY_PROCESSED	22	5	
	57174ec3-cac8-4aec-a867-7ea6ff410d3f	3	0	1009	2026-09-15	2026-09-17	PENDING	22	5	
	be3b1eda-3214-4e15-84d3-07d9177bcb4a	2	0	1009	2026-09-15	2026-09-17	PENDING	22	5	
	f4439857-8aa6-4643-b57a-e16db93c2fd2	1	0	1009	2026-09-15	2026-09-17	PENDING	22	5	
	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	